

# CROSS-MODAL BRAIN MRI SYNTHESIS VIA CONTRASTIVE DISENTANGLEMENT AND CONDITIONAL DIFFUSION

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## ABSTRACT

Cross-modal brain magnetic resonance imaging (MRI) synthesis remains challenging due to the complex interplay between anatomical structures and modality-specific contrast patterns. We propose a two-phase framework that separates anatomical and contrast representations through contrastive learning and conditional diffusion. Phase 1 uses contrastive disentanglement with dual projection heads to separate anatomy (patient-specific structure) from contrast (modality-specific appearance), reinforced by adversarial training. A learnable contrast bank stores modality prototypes independent of anatomical content. Phase 2 employs a conditional diffusion model that synthesizes target images by combining source anatomy features with target contrast codes from the bank. Training integrates anatomy-invariant and contrast-invariant contrastive losses and adversarial modality classification to enforce clear disentanglement. Evaluated on multi-modal brain MRI data (T1-w, T1ce, T2-w, FLAIR), our approach achieves one-to-many synthesis, allowing a single source modality to generate multiple target modalities without relying on source intensity patterns. The disentangled representation leads to better generalization and more realistic contrast-specific patterns than traditional entangled feature representation models.

**Index Terms**— Cross-modal synthesis, Brain MRI, Representation disentanglement, Contrastive learning, Medical image translation

## 1. INTRODUCTION

Multi-modal brain magnetic resonance imaging (MRI) captures complementary structural and functional details that are crucial for accurate clinical diagnosis and effective treatment planning. Various MRI sequences, such as T1-weighted, T1-contrast enhanced (T1ce), T2-weighted, and fluid-attenuated inversion recovery (FLAIR), capture distinct tissue characteristics and pathological variations. However, acquiring complete multi-modal MRI datasets is often impractical due to prolonged scan times, patient discomfort, and variability in clinical protocols. Cross-modal MRI synthesis addresses this challenge by generating missing modalities from available sequences. Existing approaches typically learn direct inter-modality mappings using generative adversarial networks (GANs) [1, 2] or diffusion models [3, 4]. These methods often struggle with the fundamental challenge that anatomical structure should remain consistent across modalities while contrast patterns vary significantly. Recent works in representation disentanglement [5, 6] and MRI harmonization [7] suggests that separating anatomy from modality-specific appearance can improve synthesis quality. However, current state-of-the-art methods for cross-modal MRI synthesis either learn entangled representations where anatomy and contrast are inseparable, limiting controllability and generalization or rely

on paired training with direct intensity supervision [8, 9], restricting synthesis to learned modality pairs [10], reducing practical flexibility. We introduce a two-phase framework that explicitly separates anatomical structure from modality-specific contrast through contrastive learning and conditional diffusion. Phase 1 disentangles anatomy from contrast through dual projection heads. One extracts anatomy-invariant features (patient structure) and the other extracts contrast-invariant features (modality appearance), both reinforced by adversarial training. A learnable contrast bank stores modality-specific prototypes independent of anatomical content, enabling explicit control over imaging characteristics. Phase 2 leverages a conditional diffusion model to synthesize target modality images by combining anatomy embeddings from the source image with contrast codes retrieved from the bank, guided by edge maps to preserve structural fidelity. This design enables robust one-to-many synthesis without relying on source intensity patterns, allowing a single input modality to produce multiple target sequences with distinct contrast characteristics while preserving anatomical consistency. This disentangled framework enables flexible applications: generating missing modalities when certain scans are unavailable and augmenting training datasets with synthetic multi-modal data. The main contributions of this work are:

- A **two-phase contrastive-diffusion framework** that disentangles anatomy from contrast via learnable prototypes, enabling one-to-many modality synthesis.
- **Adversarial disentanglement with contrastive learning** that separates anatomy-invariant contrast features and contrast-invariant anatomy features through dual projection heads and a learnable contrast bank.
- **Extensive experiments** on multimodal brain MRI datasets showing superior performance in one-to-many synthesis while preserving anatomical consistency across modalities.

## 2. METHODOLOGY

### 2.1. Data Preprocessing

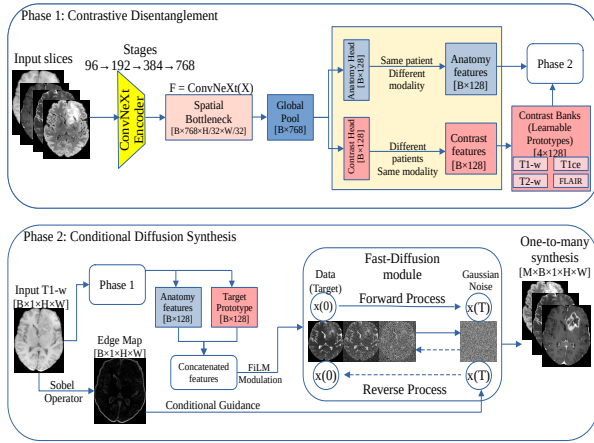
Volumes undergo min-max normalization to  $[0, 1]$  after clipping outliers at 99th percentile and resampling to  $1 \times 1 \times 1 \text{ mm}^3$ . We extract 2D axial slices ( $256 \times 256$ ) for training; volume-wise normalization maintains inter-slice consistency. Each patient contributes slices from all modalities for patient-level contrastive learning in Phase 1. For inference, slices are processed independently and reconstructed into 3D volumes; absolute intensity is recoverable via inverse normalization when needed.

### 2.2. Architecture

Our two-phase architecture is illustrated in Figure 1.

**Phase 1 (Contrastive Disentanglement):** Input images are processed by a ConvNeXt [11] encoder to extract spatial feature representations. ConvNeXt is adopted due to its modern architectural design featuring large kernels, inverted bottlenecks, and LayerNorm, which preserves convolutional locality and is well suited for anatomy-preserving feature extraction. Global average pooling produces compact feature embeddings that are passed through dual projection heads to disentangle anatomy-invariant and contrast-invariant representations. A learnable contrast bank is introduced to store modality-specific contrast prototypes.

**Phase 2 (Conditional Diffusion):** A conditional diffusion model synthesizes target-modality images by conditioning on anatomy embeddings from the source image and contrast codes retrieved from the contrast bank. Edge maps are additionally used to provide explicit structural guidance during image generation.



**Fig. 1.** The proposed two-phase framework with anatomy-contrast disentanglement. Phase 1 performs contrastive disentanglement by learning contrast-invariant anatomical embeddings and modality-specific contrast prototypes via a ConvNeXt [11] encoder. Phase 2 uses a Fast-Diffusion module conditioned on Sobel-based edge maps from the input and the disentangled embeddings to generate target modalities from a single source, enabling one-to-many synthesis.

### 2.2.1. Phase 1: Contrastive Disentanglement

Given spatial features  $\mathbf{F} \in \mathbb{R}^{B \times 768 \times H' \times W'}$  from the ConvNeXt[11] encoder, global average pooling extracts feature vectors  $\mathbf{f} \in \mathbb{R}^{B \times 768}$ . Two parallel projection heads then produce L2-normalized embeddings:

$$\mathbf{z}_a = \mathcal{N}(\text{MLP}_a(\mathbf{f})), \quad \mathbf{z}_c = \mathcal{N}(\text{MLP}_c(\mathbf{f})) \quad (1)$$

where  $\mathbf{z}_a \in \mathbb{R}^{B \times 128}$  represents anatomy-invariant features and  $\mathbf{z}_c \in \mathbb{R}^{B \times 128}$  captures contrast-invariant features, with  $\mathcal{N}(\cdot)$  denoting L2 normalization. Disentanglement is enforced through InfoNCE[12] contrastive losses. For anatomy, positive pairs are same-patient different-modality samples; for contrast, positive pairs are same-modality different-patient samples:

$$\mathcal{L}_{\text{anatomy}} = \frac{1}{B} \sum_{i=1}^B -\log \frac{\sum_{j \in \mathcal{P}_i^a} \exp(\text{sim}(\mathbf{z}_a^i, \mathbf{z}_a^j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(\mathbf{z}_a^i, \mathbf{z}_a^k)/\tau)} \quad (2)$$

$$\mathcal{L}_{\text{contrast}} = \frac{1}{B} \sum_{i=1}^B -\log \frac{\sum_{j \in \mathcal{P}_i^c} \exp(\text{sim}(\mathbf{z}_c^i, \mathbf{z}_c^j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(\mathbf{z}_c^i, \mathbf{z}_c^k)/\tau)} \quad (3)$$

where  $\mathcal{P}_i^a = \{j : p_i = p_j, m_i \neq m_j\}$  and  $\mathcal{P}_i^c = \{j : p_i \neq p_j, m_i = m_j\}$  with  $p_i$  and  $m_i$  denoting patient and modality indices.

A learnable contrast bank stores modality-specific prototypes  $\mathbf{B} \in \mathbb{R}^{M \times 128}$  where  $M = 4$  is the number of modalities. These prototypes are learned through alignment loss:

$$\mathcal{L}_{\text{bank}} = 1 - \frac{1}{B} \sum_{i=1}^B \text{sim}(\mathbf{B}_{m_i}, \mathbf{z}_c^i) \quad (4)$$

which aligns bank prototypes with image-derived contrast embeddings.

Additional regularization includes adversarial modality classification on anatomy embeddings:

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_{\mathbf{z}_a} [\text{CE}(\text{Classifier}(\text{GRL}(\mathbf{z}_a)), m)] \quad (5)$$

where GRL denotes gradient reversal layer.

Total Phase 1 loss:

$$\mathcal{L}_{\text{phase1}} = \lambda_a \mathcal{L}_{\text{anatomy}} + \lambda_c \mathcal{L}_{\text{contrast}} + \lambda_b \mathcal{L}_{\text{bank}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} \quad (6)$$

### 2.2.2. Phase 2: Conditional Diffusion Synthesis

After Phase 1, the encoder and projection heads are frozen. Fast-DDPM conditional diffusion model [13] is used to synthesize target modalities by combining anatomy embeddings from the source image with contrast codes from the contrast bank. Given a source image  $\mathbf{I}_s$  and target modality  $m_t$ , we extract anatomy features using the frozen Phase 1 encoder:

$$\mathbf{z}_a = \text{Anatomy-Head}(\text{Encoder}(\mathbf{I}_s)) \quad (7)$$

and retrieve the target contrast code from the bank:

$$\mathbf{z}_c^{\text{target}} = \mathbf{B}_{m_t} \quad (8)$$

The diffusion model operates in pixel space with forward noising process:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad (9)$$

where  $\mathbf{x}_0$  is the clean target image,  $\bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i)$ , and  $\beta_t$  follows a linear schedule from 0.0001 to 0.02.

We use a non-uniform 10-step schedule for both training and sampling, reducing computational cost. The denoising network predicts noise  $\epsilon$  conditioned on anatomy and contrast embeddings via FiLM layers which modulate features through learned scale and shift parameters [14], with additional structural guidance from edge maps  $\mathbf{E}_s = \text{Sobel}(\mathbf{I}_s)$ :

$$\epsilon_{\theta}(\mathbf{x}_t, t, \mathbf{z}_a, \mathbf{z}_c^{\text{target}}) = \text{UNet}([\mathbf{x}_t, \mathbf{E}_s], t, [\mathbf{z}_a, \mathbf{z}_c^{\text{target}}]) \quad (10)$$

During training the objective minimizes noise prediction error:

$$\mathcal{L}_{\text{diffusion}} = \mathbb{E}_{\mathbf{x}_0, \epsilon, t \sim \mathcal{T}} [\|\epsilon - \epsilon_{\theta}(\mathbf{x}_t, t, \mathbf{z}_a, \mathbf{z}_c^{\text{target}})\|_2^2] \quad (11)$$

For inference, we use deterministic DDIM sampling [15] over the same 10-step schedule, computing:

$$\mathbf{x}_{t-1} = \underbrace{\sqrt{\bar{\alpha}_{t-1}} \left( \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_{\theta}}{\sqrt{\bar{\alpha}_t}} \right)}_{\text{predicted } \mathbf{x}_0} + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_{\theta} \quad (12)$$

This design enables flexible one-to-many synthesis: given any

source modality, we generate any target modality by combining the source’s anatomy with the target’s contrast code from the bank, without relying on source intensity patterns.

### 3. EXPERIMENTS AND ANALYSIS

#### 3.1. Dataset and Training Setup

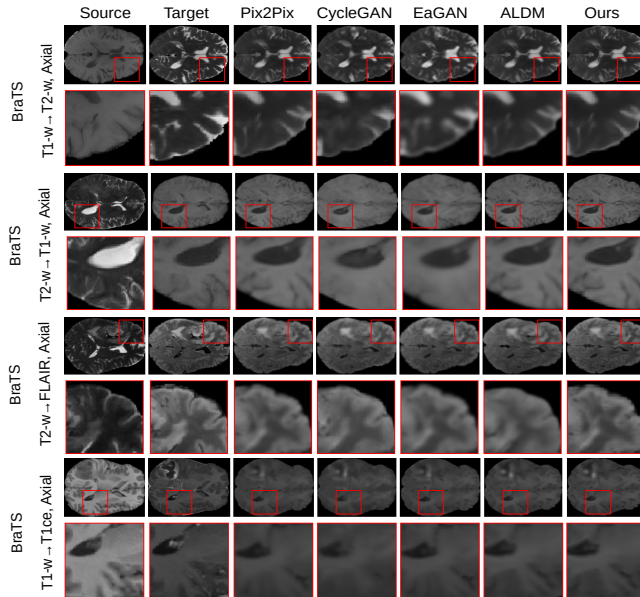
**BraTS2021** dataset [16] with 1,251 subjects (T1-w, T1ce, T2-w, FLAIR) was used for training, with 219 validation subjects for external evaluation. Additionally, **IXI** dataset [17] with 574 subjects (T1-w, T2-w, PD) validated generalizability using 459 training and 115 evaluation subjects.

**Phase 1 Training:** This was trained for 1,000 epochs using batch size of 32 patients (128 images across 4 modalities), Adam optimizer with learning rate  $3 \times 10^{-4}$ , weight decay  $1 \times 10^{-4}$ , and cosine annealing schedule. Loss weights were set to  $\lambda_a = 1.0$ ,  $\lambda_c = 1.0$ ,  $\lambda_b = 0.1$ ,  $\lambda_{adv} = 1.0$ . Temperature  $\tau = 0.07$  was used for contrastive losses.

**Phase 2 Training:** The conditional diffusion model was trained for 5,000,000 iterations using batch size 1 (all cross-modal pairs per patient), Adam optimizer ( $lr = 1 \times 10^{-4}$ ), non-uniform 10-step schedule for training and sampling, and Sobel-computed edge maps for structural guidance. Training used NVIDIA H200 GPU with ~28GB memory (reducible to ~4GB by processing one pair per iteration). Results were averaged over five experimental runs for both phases.

#### 3.2. Results and Discussion

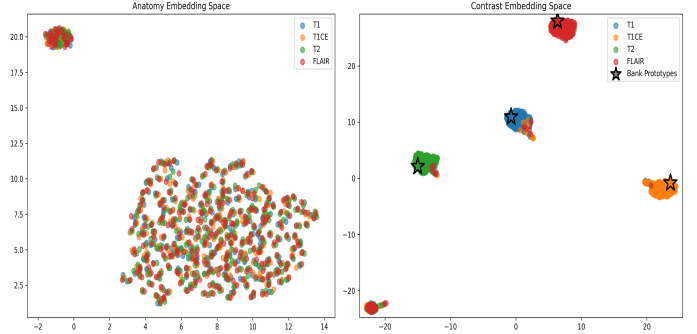
Table 1 presents image quality metrics (SSIM, PSNR, NMSE), Table 2 demonstrates one-to-many synthesis results on BraTS2021, and Figure 2 provides visual comparisons.



**Fig. 2.** Visual comparison of synthesis results on BraTS2021 dataset. Four representative examples from twelve possible cross-modal combinations are shown.

Figure 3 shows UMAP visualization of the learned embedding spaces, where the anatomy space exhibits modality mixing and the contrast space displays clear modality-specific clustering with bank prototypes near cluster centroids, demonstrating that contrastive learning effectively separates anatomy and contrast representations.

Our method achieves consistent improvements across all baselines. A single source modality can generate multiple target modalities by combining its anatomy with different contrast codes from the bank. This reduces the need for training separate models for each source-target pair, addressing practical scenarios where certain modalities may be missing or unavailable in clinical workflows.



**Fig. 3.** UMAP visualization. **Left:** Anatomy space shows modality-invariant clustering (same patient across modalities). **Right:** Contrast space shows modality-specific clustering with learnable bank prototypes (stars) at cluster centroids.

The framework enables independent optimization of encoders based on computational constraints. However, synthesis quality depends on anatomy embedding richness; source modalities poorly visualizing certain structures (e.g., T1-w weakly captures edema visible in FLAIR) limit fidelity. Additionally, the contrast bank’s dependence on training distribution may degrade performance under distribution shift. Future work could evaluate synthesized modalities in downstream segmentation tasks to assess clinical utility.

**Table 1.** Quantitative results across three tasks. The best results are shown in **bold**, second-best underlined.

| Experiment | Method   | SSIM $\uparrow$                  | PSNR $\uparrow$                   | NMSE $\downarrow$                |
|------------|----------|----------------------------------|-----------------------------------|----------------------------------|
| BraTS2021  | Pix2Pix  | 0.878 $\pm$ 0.01                 | 25.071 $\pm$ 0.81                 | 0.101 $\pm$ 0.03                 |
|            | CycleGAN | 0.813 $\pm$ 0.02                 | 23.710 $\pm$ 1.23                 | 0.161 $\pm$ 0.05                 |
|            | Ea-GAN   | 0.880 $\pm$ 0.01                 | 24.872 $\pm$ 0.91                 | 0.098 $\pm$ 0.02                 |
|            | ALDM     | 0.901 $\pm$ 0.01                 | 25.751 $\pm$ 0.87                 | 0.082 $\pm$ 0.01                 |
|            | Ours     | <b>0.921<math>\pm</math>0.01</b> | <b>26.872<math>\pm</math>0.69</b> | <b>0.076<math>\pm</math>0.01</b> |
| BraTS2021  | Pix2Pix  | 0.847 $\pm$ 0.02                 | 24.223 $\pm$ 1.33                 | 0.122 $\pm$ 0.03                 |
|            | CycleGAN | 0.811 $\pm$ 0.03                 | 23.118 $\pm$ 1.41                 | 0.168 $\pm$ 0.04                 |
|            | Ea-GAN   | 0.849 $\pm$ 0.03                 | 24.271 $\pm$ 1.19                 | 0.119 $\pm$ 0.03                 |
|            | ALDM     | 0.862 $\pm$ 0.02                 | 25.018 $\pm$ 1.10                 | 0.101 $\pm$ 0.02                 |
|            | Ours     | <b>0.875<math>\pm</math>0.02</b> | <b>25.968<math>\pm</math>1.01</b> | <b>0.100<math>\pm</math>0.01</b> |
| IXI        | Pix2Pix  | 0.887 $\pm$ 0.02                 | 26.116 $\pm$ 0.83                 | 0.095 $\pm$ 0.03                 |
|            | CycleGAN | 0.878 $\pm$ 0.03                 | 25.105 $\pm$ 1.21                 | 0.101 $\pm$ 0.04                 |
|            | Ea-GAN   | 0.901 $\pm$ 0.03                 | 26.461 $\pm$ 0.88                 | 0.089 $\pm$ 0.03                 |
|            | ALDM     | 0.919 $\pm$ 0.02                 | 27.561 $\pm$ 0.85                 | 0.073 $\pm$ 0.03                 |
|            | Ours     | <b>0.925<math>\pm</math>0.02</b> | <b>27.998<math>\pm</math>0.80</b> | <b>0.070<math>\pm</math>0.03</b> |

Multi-source conditioning could provide richer anatomical information (e.g., jointly encoding T1-w and T2-w may improve FLAIR synthesis), and hierarchical contrast banks with spatial prototypes could model complex intensity patterns. An ablation study

is done using a leave-one-out strategy. For analysis, we focus on the T1-w-to-PD translation task using the IXI dataset. The results are summarized in Table 3.

**Table 2.** Cross-modality translation results on BraTS2021. Values show mean $\pm$ std for SSIM ( $\uparrow$ ), PSNR ( $\uparrow$ ), NMSE ( $\downarrow$ ).

| Source | Target | SSIM $\uparrow$  | PSNR $\uparrow$   | NMSE $\downarrow$ |
|--------|--------|------------------|-------------------|-------------------|
| T1-w   | T1ce   | 0.914 $\pm$ 0.02 | 26.521 $\pm$ 0.80 | 0.077 $\pm$ 0.02  |
| T1-w   | T2-w   | 0.921 $\pm$ 0.01 | 26.872 $\pm$ 0.69 | 0.076 $\pm$ 0.01  |
| T1-w   | FLAIR  | 0.862 $\pm$ 0.02 | 25.010 $\pm$ 0.72 | 0.101 $\pm$ 0.03  |
| T1ce   | T1-w   | 0.931 $\pm$ 0.03 | 26.491 $\pm$ 0.80 | 0.072 $\pm$ 0.02  |
| T1ce   | T2-w   | 0.912 $\pm$ 0.01 | 26.113 $\pm$ 0.86 | 0.090 $\pm$ 0.03  |
| T1ce   | FLAIR  | 0.885 $\pm$ 0.03 | 25.784 $\pm$ 0.81 | 0.089 $\pm$ 0.05  |
| T2-w   | T1-w   | 0.912 $\pm$ 0.02 | 25.910 $\pm$ 0.84 | 0.081 $\pm$ 0.02  |
| T2-w   | T1ce   | 0.903 $\pm$ 0.02 | 25.784 $\pm$ 1.10 | 0.082 $\pm$ 0.03  |
| T2-w   | FLAIR  | 0.875 $\pm$ 0.02 | 25.968 $\pm$ 1.01 | 0.100 $\pm$ 0.01  |
| FLAIR  | T1-w   | 0.917 $\pm$ 0.04 | 25.913 $\pm$ 0.74 | 0.085 $\pm$ 0.02  |
| FLAIR  | T1ce   | 0.912 $\pm$ 0.02 | 26.103 $\pm$ 1.01 | 0.091 $\pm$ 0.02  |
| FLAIR  | T2-w   | 0.952 $\pm$ 0.02 | 26.889 $\pm$ 0.61 | 0.083 $\pm$ 0.02  |

**Table 3.** Ablation study systematically removing Edge Map (Sobel-based structural guidance), Anatomy Embedding (source structure features from Phase 1), Contrast Prototype (target modality codes from learnable bank), and FiLM (feature modulation layers) on T1-w-to-PD translation task on IXI dataset.

| Edge Map | Model Components  |                    |      | Image Quality Metrics            |                                   |                                  |
|----------|-------------------|--------------------|------|----------------------------------|-----------------------------------|----------------------------------|
|          | Anatomy Embedding | Contrast Prototype | FiLM | SSIM $\uparrow$                  | PSNR $\uparrow$                   | NMSE $\downarrow$                |
| ✓        |                   |                    |      | 0.883 $\pm$ 0.02                 | 26.512 $\pm$ 0.85                 | 0.081 $\pm$ 0.02                 |
| ✓        | ✓                 |                    |      | 0.887 $\pm$ 0.04                 | 26.881 $\pm$ 0.83                 | 0.078 $\pm$ 0.03                 |
| ✓        | ✓                 | ✓                  |      | 0.899 $\pm$ 0.03                 | 27.112 $\pm$ 0.85                 | 0.074 $\pm$ 0.03                 |
| ✓        | ✓                 | ✓                  | ✓    | <b>0.925<math>\pm</math>0.02</b> | <b>27.998<math>\pm</math>0.80</b> | <b>0.070<math>\pm</math>0.03</b> |

#### 4. CONCLUSION

In this work, we proposed a two-phase framework for cross-modal brain MRI synthesis that explicitly disentangles anatomical structure from modality-specific contrast. By separating representation learning from image generation, our approach enables one-to-many synthesis where any source modality can generate any target modality without relying on intensity-based translation. Through comprehensive evaluation on BraTS2021 and IXI datasets, we demonstrate superior synthesis quality compared to existing methods. This framework could enhance clinical workflows by generating missing modalities under acquisition constraints, reducing scan time while maintaining diagnostic quality. The design provides a principled approach to cross-modal synthesis across diverse anatomy-contrast combinations.

#### 5. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by BraTS2021[16] and IXI[17]. Ethical approval was not required as confirmed by the license attached with the open access data.

#### 6. ACKNOWLEDGMENTS

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